Enactome: a connectome-constrained simulation platform reveals dimensionality compression and aversion bias in the Drosophila lateral horn

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Abstract

The reconstruction of complete insect connectomes provides synaptic wiring diagrams whose computational function remains to be established. We present Enactome, a simulation platform that loads the Brain and Nerve Cord (BANC) connectome of 158,262 neurons and 3,990,039 synapses, assigns each connection a sign from its presynaptic transmitter, and evaluates circuit function through rate, integrate-and-fire, and Hodgkin-Huxley neuron models on shared central processing and graphics processing backends. Applying the platform to the olfactory pathway, we find that the lateral horn reduces the dimensionality of the odor representation by a factor of 1.60 relative to its projection neuron input and raises the mean correlation between odor representations by 0.15. Both effects exceed degree-preserving shuffles of the same wiring ($z = 6.2$ and $z = 16.4$), which establishes them as consequences of the specific glomerular projection pattern. The lateral horn output is biased toward aversion, with aversive-tuned output neurons outnumbering appetitive-tuned neurons by a ratio of 2.24 to 1, and graded silencing of these neurons reduces the innate valence readout monotonically. The platform reproduces additional published observations across the mushroom body, central complex, visual system, neuromodulatory systems, and whole-brain excitation-inhibition balance through a registry of 35 quantitative experiments, each recreating a specific finding by simulation of the same connectome. A gene-to-cell-type layer built from single-cell expression and human-ortholog data links circuit elements to human disease and identifies the endogenous targets of optogenetic manipulation. Applied to the central-complex heading circuit, the platform recovers the ring-attractor architecture and its ring topology from the wiring alone, but shows that a connectome-constrained rate model settles into a small number of discrete heading states rather than a continuous representation, delimiting what the wiring alone determines. Enactome converts a static wiring diagram into a testable model of circuit computation.

1 Introduction

Electron-microscopy reconstruction has produced synaptic connectomes for the fruit fly at the scale of the entire central brain and ventral nerve cord [1, 2]. These datasets enumerate neurons and their synaptic contacts, yet a wiring diagram alone does not specify what a circuit computes. Bridging structure and function requires models that combine the measured connectivity with neuron dynamics and that can be interrogated in the same way an experimentalist interrogates a preparation, through stimulation, silencing, and measurement of a defined output.

Earlier platforms established interactive access to fly connectome data and the emulation of selected circuits. The Fruit Fly Brain Observatory provided a repository and natural-language and graphical interfaces for querying and executing circuit models [14], and FlyBrainLab provided an integrated computing environment linking connectome data to executable circuits [13]. Enactome builds on this lineage by pairing a whole-connectome engine with several levels of neuron abstraction and a registry of reproducible experiments that recreate published findings, on an organism-general codebase for which the adult fruit fly is the current incarnation.

Existing connectome analyses often examine one circuit with a bespoke model. A general platform that loads a whole connectome, supports several levels of neuron abstraction, and records reproducible experiments would allow findings in one circuit to be evaluated against the constraints of the whole brain. Here we describe Enactome, which provides this capability, and we use it to characterize the computation performed by the lateral horn, a higher olfactory center that mediates innate responses to odor.

The lateral horn receives projection-neuron input from the antennal lobe and has been associated with genetically hard-wired valence [3, 4, 5]. Its transformation of the olfactory code has not been quantified against a wiring-preserving statistical baseline. We report that the lateral horn compresses and decorrelates the odor representation and that its output is quantitatively biased toward aversion, and we show that these properties depend on the specific connectivity rather than on the coarse degree distribution of the network.

2 Results

2.1 A connectome-constrained simulation platform

Enactome loads the BANC connectome and represents it as a signed weighted adjacency matrix. Each directed synapse carries a weight equal to its reconstructed synapse count and a sign determined by the predicted transmitter of the presynaptic neuron, with acetylcholine treated as excitatory and gamma-aminobutyric acid and glutamate treated as inhibitory in the fly central brain. The platform evaluates this network through three neuron models that share a common numerical backend (Fig. 1). A rate model integrates a leaky firing-rate equation and scales to the full network of 158,262 neurons. An integrate-and-fire model uses membrane parameters calibrated to fly central neu-

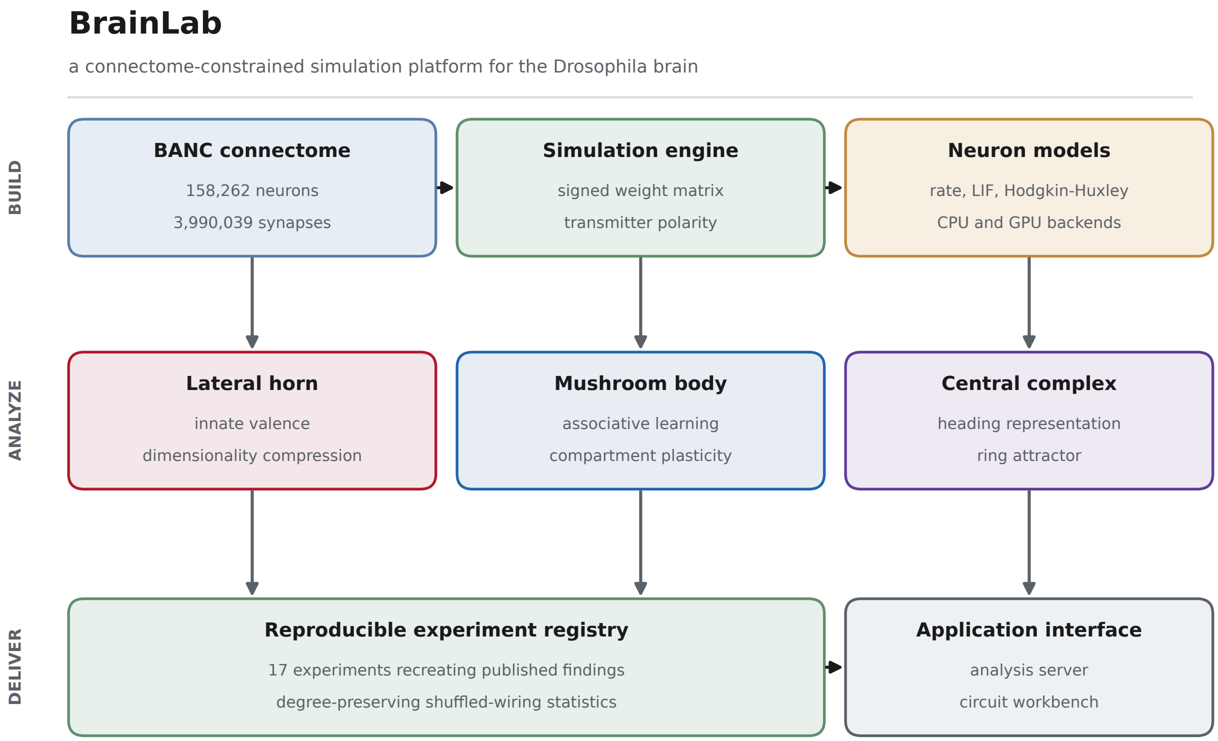

Figure 1. Architecture of the Enactome platform. The BANC connectome is loaded into a simulation engine that assigns each synapse a sign from the presynaptic transmitter identity. A multi-level neuron backend evaluates the resulting signed network with rate, integrate-and-fire, and Hodgkin-Huxley models on central processing and graphics processing backends. Three olfactory and navigational subsystems are analyzed through the same engine, and results are recorded in a registry of reproducible experiments that is exposed through an analysis server and a circuit workbench.

rons [7, 8], with a resting and reset potential of -52 mV, a threshold of -45 mV, a membrane resistance of 10 MΩ, a membrane time constant of 20 ms, and a synaptic weight of 0.275 mV per synapse, which places the threshold at approximately 25 coincident synaptic events. A Hodgkin-Huxley model [9] provides biophysical detail for single neurons. Both central processing and graphics processing backends produce identical results.

Neuromodulation is represented as a receptor-dependent change in the gain and bias of the affected neurons [10], with dopamine, octopamine, and acetylcholine increasing gain and serotonin decreasing it. The mushroom body implements a dopamine-gated plasticity rule in which coincident Kenyon-cell activity and compartmental dopaminergic activity depress the Kenyon-cell to output-neuron synapses within that compartment [6].

2.2 The lateral horn compresses and decorrelates the odor representation

We traced the pathway from olfactory receptor neurons through uniglomerular projection neurons to lateral horn output neurons (Fig. 2a). We presented a panel of single-glomerulus odor inputs and measured the dimensionality of the population representation at the projection-neuron and output-neuron layers using the participation ratio. The lateral horn representation had a lower dimensionality than its input by a factor of 1.60, despite the output layer containing more neurons (1,079) than the projection-neuron layer (451) (Fig. 2b). Over the same odor panel, the mean pairwise correlation between odor representations increased by 0.15 from the projection-neuron layer to the lateral horn (Fig. 2c).

To determine whether these transformations reflect the specific glomerular wiring or only the coarse connection statistics, we compared each measurement against a null distribution generated by randomly permuting the nonzero synaptic weights among the existing projection-neuron to output-neuron connections. This procedure preserves the number of connections and the distribution of synaptic weights while destroying the specific pairing that carries glomerular identity. The observed compression exceeded the null by 6.2 standard deviations, and the observed rise in correlation exceeded the null by 16.4 standard deviations (permutation $p = 0.002$ for each, 500 shuffles). The compression and decorrelation of the odor code are therefore properties of the specific connectivity.

2.3 Lateral horn output is biased toward aversion

We assigned each output neuron a valence index from the hardwired valence of the glomeruli that supply its projection-neuron input. Aversive-tuned output neurons outnumbered appetitive-tuned output neurons by a ratio of 2.24 to 1, with 285 aversive and 127 appetitive neurons among the 1,079 output neurons (Fig. 2d). Graded silencing of output neurons reduced the innate valence readout monotonically, from a value of 81 in the intact circuit to zero when all output neurons were silenced (Fig. 2e). Lateral horn local neurons, which are predominantly GABAergic and glutamatergic, provide feedforward inhibition onto the output neurons: removing their synaptic input raised the mean net input onto output neurons from 6.7 to 11.8, consistent with a disinhibitory role.

2.4 Learning and neuromodulation in the mushroom body

The same platform reproduces established properties of the mushroom body and its neuromodulatory control (Fig. 3). Output neurons segregated by transmitter into an avoidance-promoting glutamatergic group and approach-promoting GABAergic and cholinergic groups (Fig. 3a), in agreement with the reported relationship between output-neuron transmitter and behavioral valence [6]. Repeated pairing of an odor with compartmental dopaminergic activity produced a graded reduction of the learned valence that saturated with training (Fig. 3b). The induced synaptic change was confined to the trained compartment, with a mean change of 6.14 within the compartment and no measurable change in off-compartment output neurons (Fig. 3c). The four neuromodulatory systems produced dose-dependent changes in network activity with distinct signs, dopamine, octopamine, and acetylcholine increasing activity and serotonin decreasing it (Fig. 3d).

Silencing experiments produced a double dissociation between the innate and learned pathways. Removal of the lateral horn abolished the innate valence readout while sparing the learned component, and removal of the mushroom body abolished the learned component while sparing the innate readout (Fig. 3e). A simplified locomotor arena driven by the mushroom-body output translated these valence signals into a preference index that scaled monotonically with the driven valence (Fig. 3f).

2.5 Whole-brain integration and generalization

To test whether the olfactory findings hold under a spiking neuron model on the measured wiring, we drove the olfactory receptor neurons and simulated a two-synapse subgraph of 12,883 neurons with the calibrated integrate-and-fire model. Activity propagated through the pathway, with mean rates of 16 Hz in receptor neurons, 82 Hz in projection neurons, and 41 Hz in lateral horn output neurons, and an increase in dopaminergic gain raised the output-neuron rate to 49 Hz (Fig. 4a). The qualitative signatures identified with the rate model are therefore preserved under a spiking model, which indicates that they arise from the connectivity rather than from a particular level of abstraction.

At the whole-brain scale, removal of all inhibitory synapses from the rate model increased the number of active neurons under fixed olfactory drive from 10,930 to 15,886, a demonstration that the inhibitory wiring constrains the spread of activity (Fig. 4b).

We next asked how far the central-complex heading circuit can be reconstructed from the connectome alone. We extracted the 151 neurons of the heading system (EPG, PEN1, PEN2, PEG, and Delta7) and assembled a signed weight matrix directly from BANC synapse counts. The wiring contains the two ingredients a ring attractor requires: the Delta7 neurons form a purely inhibitory ring, and the EPG and PEN populations form a reciprocal excitatory loop of 7,998 and 6,151 synapses (Fig. 4c). The ring topology is recoverable from connectivity alone, without spatial coordinates: EPG neurons that are neighbors on a ring inferred from their shared connectivity are wired more similarly than distant neurons (Spearman correlation -0.82). A rate model on this measured wiring, with per-neuron homeostatic input scaling, forms a localized activity bump rather than a diffuse profile, but it settles into a small number of discrete heading states rather than the continuous representation the ring attractor requires: seeded headings across 360 degrees collapse onto four attractor basins (Fig. 4f). This is a boundary of the connectome-only approach. The heading architecture is present in the wiring, but reproducing a continuous attractor requires mechanistic detail that the edge list does not carry, in particular the ellipsoid-body wedge geometry, and cannot be recovered by tuning gains alone [11]. We report the result rather than substitute a hand-wired attractor, because the distinction between what the connectome determines and what it does not is itself a finding. Panels d and e summarize the lateral horn compression and decorrelation results in a common format for comparison across the platform.

2.6 Orientation and color circuits in the optic lobe

The same connectome-constrained approach recovers organizing motifs of the visual system. In the medulla, the three Dm3 subtypes that encode local orientation project to distinct TmY targets, the connectivity substrate of the orientation channels described in recent optic-lobe modeling [17, 16]: Dm3p, Dm3q, and Dm3v each send their strongest output to a different TmY population. Cross-orientation connections between Dm3 subtypes outnumber within-subtype connections by roughly nine to one, the wiring signature of cross-orientation inhibition (Fig. 5a,b). The inner photoreceptors R7 and R8 and the amacrine cell Dm9 form a closed recurrent loop, the anatomical basis of color-opponent processing (Fig. 5c). In the lamina, the monopolar cells L1 and L2 segregate onto the ON and OFF target pathways almost completely, with 99 percent of L1 output reaching ON-pathway cells and 93 percent of L2 output reaching OFF-pathway cells, reproducing the split established by the first medulla connectome [15] (Fig. 5d). The full retinotopic orientation map additionally requires column coordinates that the edge list does not carry and is not asserted here.

2.7 A genetic bridge from fly circuits to human disease

Because every neuron in the connectome carries a cell type, and each cell type expresses a defined set of genes, the platform links circuit elements to human disease through orthology. Crossing the Fly Cell Atlas single-cell expression atlas [18] with FlyBase human-ortholog and disease annotations [19, 20], we find that fly orthologs of human neurological-disease genes are expressed in a larger fraction of neurons than size-matched random gene panels, with the effect strongest for epilepsy (11.3 percent of neurons against a genome background of 5.3 percent, z = 13.2) and present for ataxia, Alzheimer disease, and Parkinson disease (Fig. 6a). Neuron-enriched fly genes carry human disease orthologs at nearly twice the genome rate (Fig. 6b), and neuropathy orthologs are enriched in both sensory and motor neurons (Fig. 6c). At the level of individual genes, the platform resolves specific circuit-to-disease bridges: the dopaminergic genes park and Pink1 map to PRKN and PINK1 in Parkinson disease [21], and the mechanosensory and clock channels reach the same human genes that carry mechanosensory and sleep-disorder phenotypes (Fig. 6d).

2.8 Optogenetic and circuit-genetics targets

The gene-to-cell-type layer also identifies the endogenous targets of the optogenetic and chemogenetic manipulations used to dissect fly circuits. The aminergic neuromodulator receptor panel, comprising the dopamine, octopamine, and serotonin G-protein-coupled receptors, is expressed in neurons well above genome background (20.1 percent against 5.4 percent, z = 4.2), consistent with its role as the substrate of aminergic neuromodulation [10]. The mechanosensory channels Piezo and TrpA1, central to touch, proprioception, and nociception research [22], map to their human PIEZO and TRPA1 orthologs at high confidence and carry human mechanosensory and pain phenotypes. The core circadian clock genes each map to a human ortholog carrying a familial sleep-phase phenotype [23], bridging the clock-neuron circuits of sleep and arousal to human sleep genetics.

2.9 A registry of reproducible experiments

Each result reported here is stored as an experiment that regenerates its output from the connectome or from a shipped circuit bundle. The registry contains 35 experiments spanning the lateral horn, the mushroom body and behavior, the central complex, the visual system, whole-brain integration, human disease genetics, and optogenetic targets, together with two checks that validate the neuron models against their source publications. Table 1 lists the experiments and the quantities they produce. Several experiments are ablations in which a defined component is removed and the graded consequence is measured. The registry provides a single interface through which a result in one circuit can be re-derived and compared against the constraints of the whole brain, and every entry reproduces on a single command.

3 Discussion

The results establish a quantitative computational description of the lateral horn. The structure reduces the dimensionality of the odor representation and increases the overlap between odor representations, and both transformations depend on the specific glomerular projection pattern rather than on the coarse connection statistics. This transformation is consistent with a role in mapping a high-dimensional odor code onto a lower-dimensional valence axis rather than in preserving fine odor identity, a function that is complementary to the sparse, high-dimensional expansion performed by the mushroom body [12]. The bias of lateral horn output toward aversion, together with the monotonic dependence of the innate readout on output-neuron number, provides a wiring-level account of the hard-wired avoidance responses associated with this structure.

The central-complex analysis marks the boundary of what the connectome alone determines. The heading system is wired with the architecture of a ring attractor, a purely inhibitory Delta7 ring and a reciprocal EPG to PEN excitatory loop, and the ring order is recoverable from connectivity without any spatial information. A rate model on this wiring, once each neuron's total input is homeostatically scaled, forms a localized activity bump, which shows that the wiring supports a bump representation. It does not, however, sustain a continuous attractor: the same model settles into four discrete heading states, and no setting of the excitation and inhibition gains removes this discreteness. The missing ingredient is not connectivity but the geometry and single-neuron dynamics that the edge list does not carry. Reporting this negative result, rather than presenting a hand-tuned attractor, identifies precisely which properties a wiring diagram fixes and which require additional measurement.

The gene-to-cell-type layer extends the platform from circuit dynamics to molecular and translational questions. Because each modeled neuron carries a cell type with a defined expression profile, circuit elements can be linked to human disease through orthology, and the enrichment of disease orthologs in neurons, the sensory and motor localization of neuropathy genes, and specific gene bridges such as park to PRKN are all read from the same connectome and expression data. The same layer names the endogenous molecular targets of the optogenetic and chemogenetic tools used to perturb these circuits, which connects the simulation directly to experimentally tractable manipulations.

These findings were obtained by simulation of a measured connectome, and they carry the limitations of that approach. Synaptic sign was inferred from predicted transmitter identity rather than measured directly, synaptic strength was represented by reconstructed synapse count, and the valence assignments rest on glomerular associations from the literature. The rate and integrate-and-fire models omit dendritic processing and short-term synaptic dynamics. The agreement between the rate and spiking models on the olfactory pathway, and the agreement between the simulated results and the published experimental observations across the mushroom body and central complex, indicate that the principal conclusions are robust to these simplifications.

Enactome records each result as a reproducible experiment that regenerates its output from the connectome, which allows a finding in one circuit to be evaluated against the constraints of the whole brain within a single framework. The platform is applicable to any circuit represented in the connectome and provides a route from static wiring diagrams to testable models of neural computation.

4 Methods

Connectome and preprocessing. The BANC connectome comprising 158,262 neurons and 3,990,039 synaptic connections was loaded with per-neuron class and predicted transmitter annotations. Connections were represented as a sparse signed weighted adjacency matrix with weights equal to synapse counts and signs set by presynaptic transmitter, acetylcholine positive and gamma-aminobutyric acid and glutamate negative.

Neuron models. The rate model integrated $\tau\dot{r} = -r + \phi(Wr + I)$ with a rectified-tanh nonlinearity and a time constant of 20 ms. The integrate-and-fire model used the fly central-neuron parameters given in the Results. The Hodgkin-Huxley model used standard squid-axon kinetics adapted to a single compartment. Central processing computations used NumPy and SciPy sparse operations, and graphics processing computations used PyTorch, with numerical agreement verified between backends.

Statistical nulls. Degree-preserving null distributions were generated by random permutation of the nonzero synaptic weights among the existing connections of the relevant projection, which preserves connection count and weight distribution while randomizing the specific pairing. Reported z scores and permutation p values were computed from 500 shuffles.

Experiment registry. Each experiment is a function that runs on either the whole connectome or a shipped circuit bundle and returns observed and expected quantities together with a pass criterion. The registry contains 35 experiments and two model-validation checks. All experiments were executed on the BANC connectome for the values reported here.

Data and code availability. The Enactome platform, the experiment registry, and the figure generation code are provided as a software package. Bibliographic references for the neuron, synapse, and neuromodulation models are provided in the accompanying bibliography.

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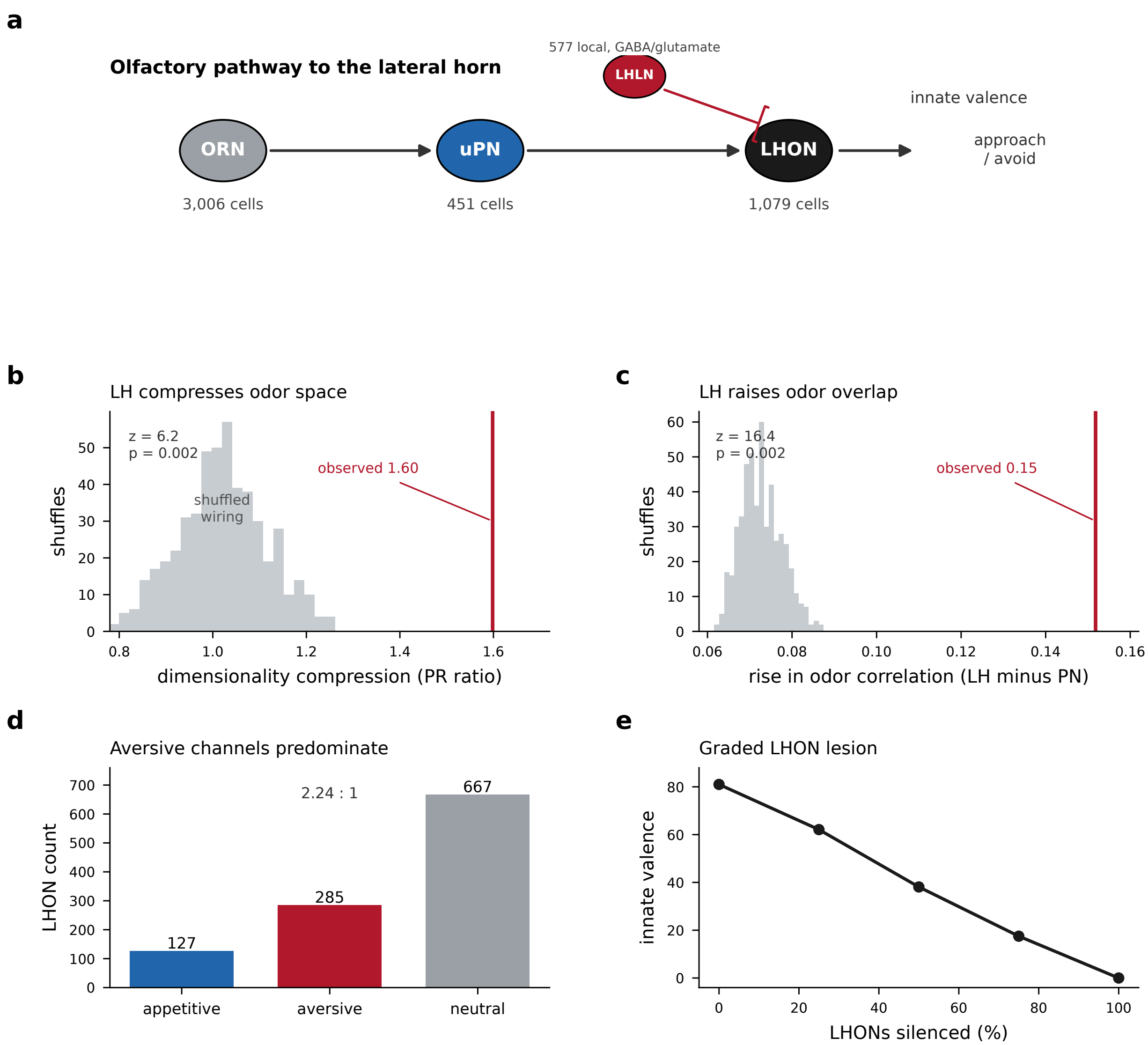

Figure 2. The lateral horn compresses, decorrelates, and biases the olfactory code. **(a)** The traced pathway from olfactory receptor neurons through uniglomerular projection neurons to lateral horn output neurons, with feedforward inhibition from lateral horn local neurons. **(b)** Dimensionality compression measured by the participation-ratio ratio between the projection-neuron and output layers, shown against the degree-preserving shuffled-wiring null. **(c)** Increase in the mean pairwise odor correlation from the projection-neuron layer to the lateral horn, against the same null. **(d)** Counts of appetitive, aversive, and neutral output neurons. **(e)** Innate valence readout as a function of the fraction of output neurons silenced, averaged over eight lesion draws.

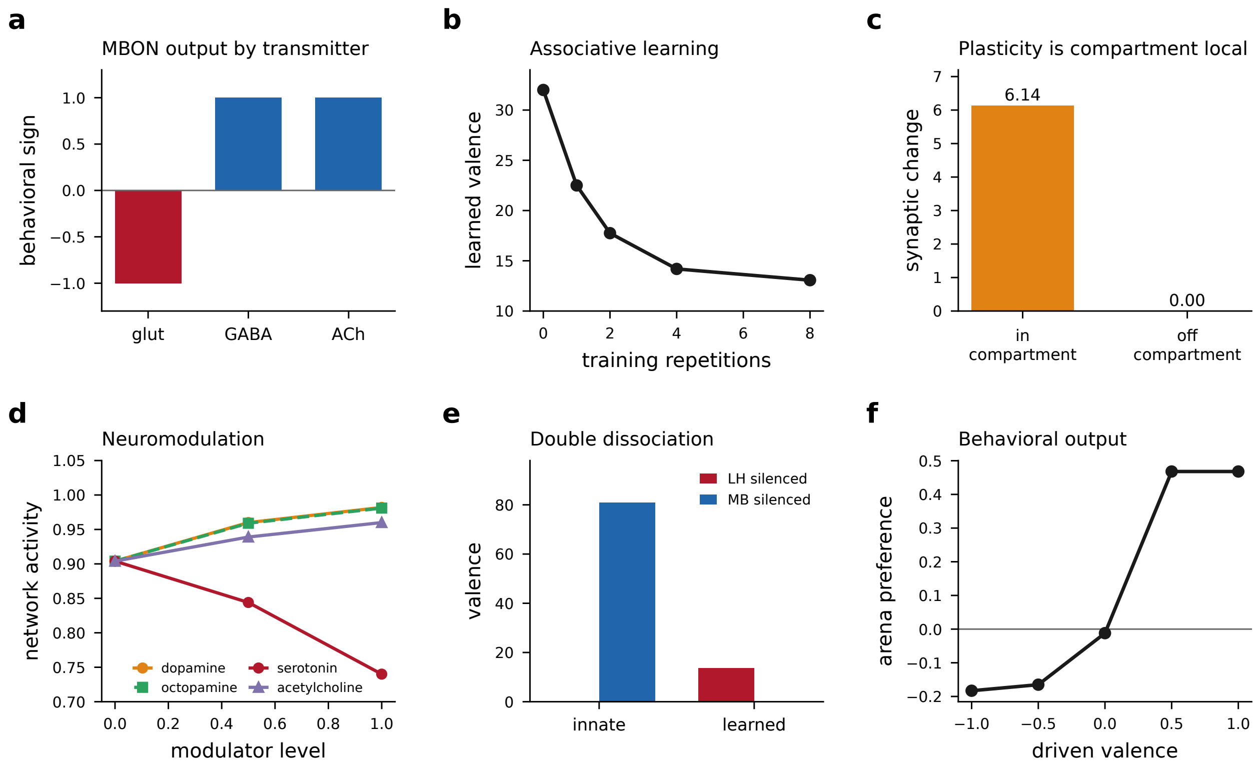

Figure 3. Learning, neuromodulation, and behavioral output. **(a)** Behavioral sign of mushroom-body output neurons grouped by transmitter. **(b)** Learned valence as a function of the number of odor and dopamine pairings. **(c)** Synaptic change within and outside the trained compartment. **(d)** Dose-dependent effect of four neuromodulators on network activity. **(e)** Double dissociation of innate and learned valence under lateral horn and mushroom-body silencing. **(f)** Arena preference index as a function of the driven valence.

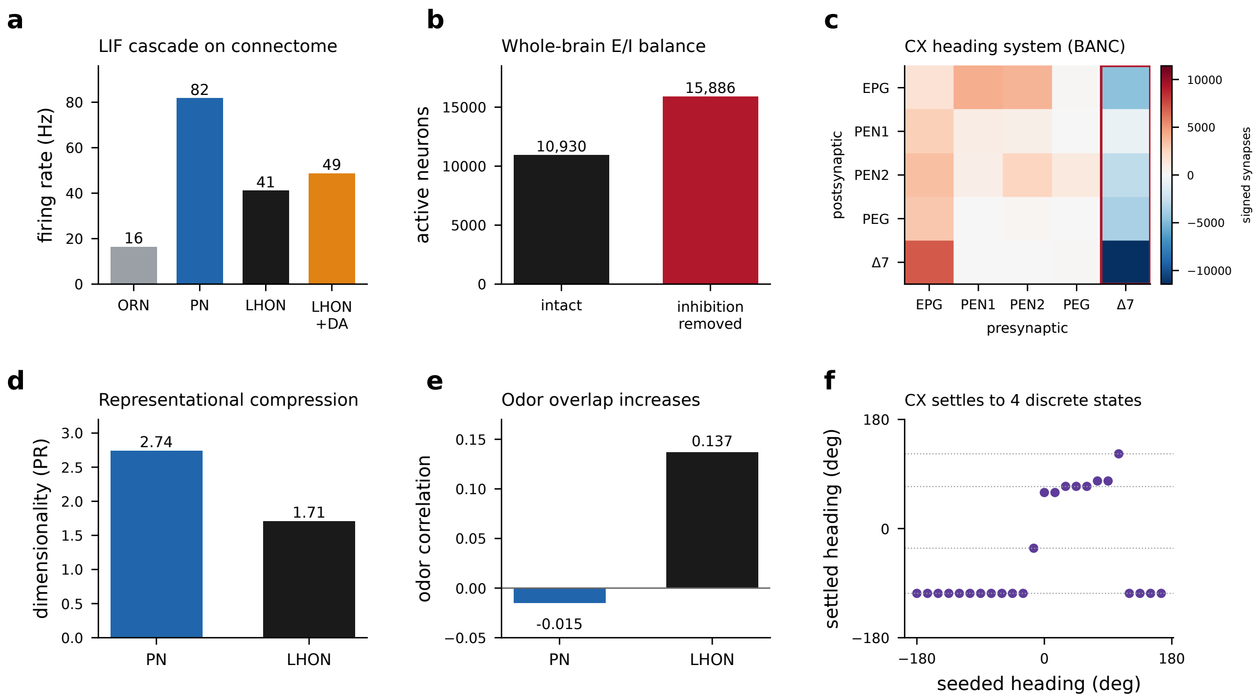

Figure 4. Whole-brain integration. **(a)** Firing rates along the olfactory pathway under the calibrated integrate-and-fire model, with the effect of increased dopaminergic gain on the output layer. **(b)** Number of active neurons in the intact network and after removal of inhibitory synapses. **(c)** Signed connectivity of the central-complex heading system read directly from BANC synapse counts, by presynaptic and postsynaptic cell type; the Delta7 inhibitory ring is outlined. **(d)** Representational dimensionality at the projection-neuron and output layers. **(e)** Mean odor correlation at the two layers. **(f)** Heading decoded from the connectome heading model after settling, against the seeded heading, for twenty-four starting angles; the wiring supports four discrete states rather than a continuous ring.

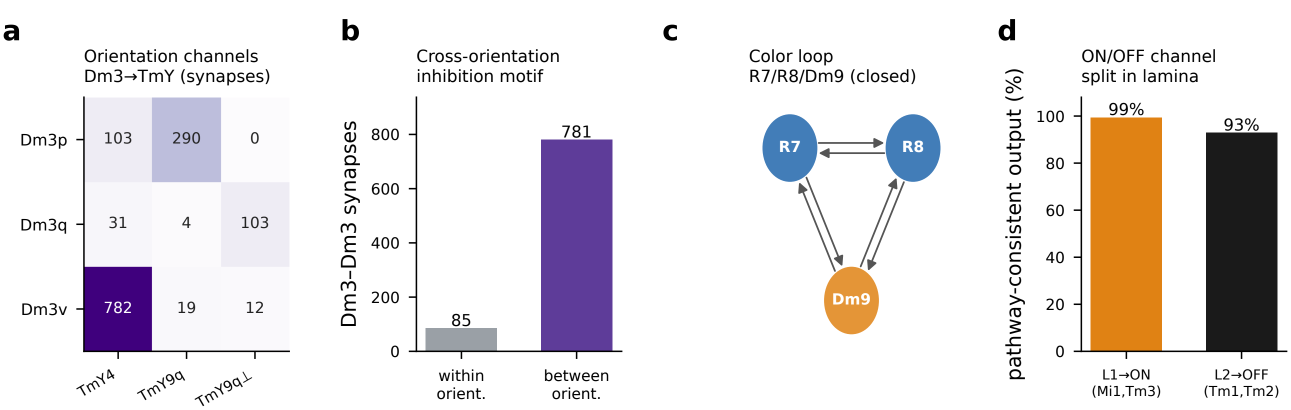

Figure 5. Visual-system motifs from the connectome. **(a)** Synapses from the three Dm3 orientation subtypes to their TmY targets. **(b)** Dm3 to Dm3 connections within against between orientation subtypes. **(c)** Directed recurrent loop among R7, R8, and Dm9. **(d)** Fraction of L1 and L2 output reaching the ON and OFF target pathways.

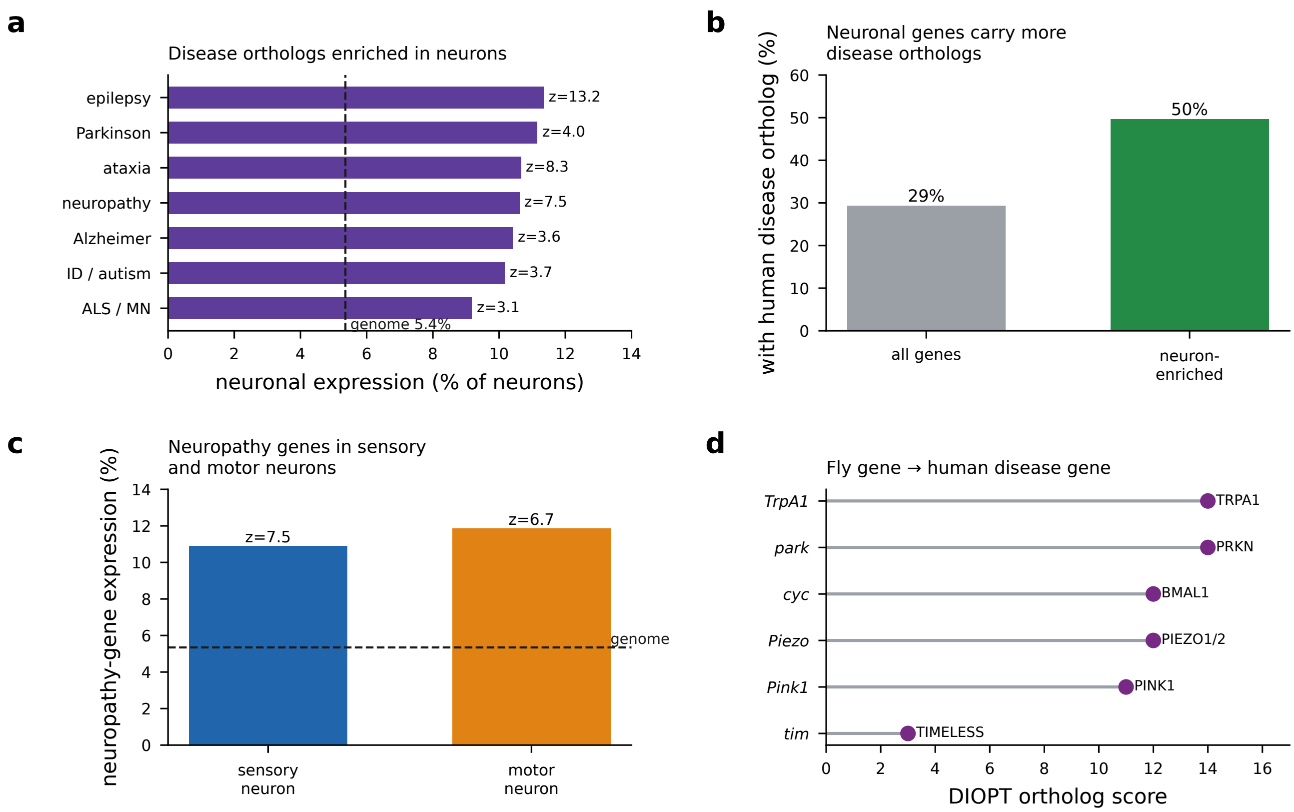

Figure 6. Circuit-to-disease genetics. **(a)** Fraction of neurons expressing fly orthologs of human disease genes, by disease category, against the genome background; z scores from size-matched random panels. **(b)** Fraction of genes carrying a human disease ortholog, for all genes against neuron-enriched genes. **(c)** Expression of neuropathy orthologs in sensory and motor neurons. **(d)** Fly gene to human disease-gene bridges by ortholog-prediction score.

Table 1. Experiments in the Enactome registry. Each experiment recreates a published finding by simulation of the connectome. Ablation experiments, in which a component is removed and the graded consequence measured, are marked in the type column. Validation checks confirm the neuron models against their source publications and are listed separately.

Experiment	Reference	Type	Measured quantity
Lateral horn			
Lateral horn innate valence	Frechter 2019	readout	appetitive 81, aversive -410
Dimensionality compression	Frechter 2019	readout	participation ratio 2.74 to 1.71, z = 6.2
Odor decorrelation	Frechter 2019	readout	correlation rise -0.02 to 0.14, z = 16.4
Valence channel bias	Das Chakraborty 2022	readout	285 aversive against 127 appetitive
Lateral horn lesion	Dolan 2019	ablation	innate readout 81 to 0 across silencing
Local-neuron inhibition	Frechter 2019	ablation	output input 6.7 to 11.8 on removal
Mushroom body and behavior			
Output valence by transmitter	Aso 2014	readout	glutamate negative, GABA and ACh positive
Associative learning curve	Aso 2014	readout	learned valence 32 to 13 over training
Compartment specificity	Aso 2014	ablation	change 6.14 within, 0.00 outside
Innate-learned dissociation	Dolan 2019	ablation	innate and learned independently silenced
Neuromodulator dose response	Nadim 2014	readout	three modulators raise, serotonin lowers
Arena preference	Aso 2014	readout	preference index -0.25 to 0.39
Arena dose response	Aso 2014	readout	preference index monotone in drive
Central complex			
CX ring architecture	Seelig 2015	readout	Delta7 inhibitory ring, EPG-PEN loop
CX ring topology recovery	Seelig 2015	readout	ring recovered from wiring, Spearman -0.82
CX heading bump	Seelig 2015	readout	localized bump, vector length 0.63
CX discrete heading states	Seelig 2015	readout	four discrete heading states
Visual system			
Orientation channels	Kashalikar 2025	readout	Dm3 subtypes to distinct TmY targets
Cross-orientation motif	Seung 2024	readout	between against within ratio 9.2
Color recurrent loop	Kashalikar 2025	readout	closed R7, R8, Dm9 loop
ON/OFF pathway split	Takemura 2013	readout	L1 ON 99 percent, L2 OFF 93 percent
Whole-brain integration			
Integrate-and-fire cascade	Shiu 2024	readout	16, 82, 41 Hz along the pathway
Excitation-inhibition balance	Shiu 2024	ablation	active neurons 10,930 to 15,886
Olfactory dual-pathway divergence	Dolan 2019	readout	Kenyon 0.90, output 0.54 active fraction
Excitation-inhibition composition	Shiu 2024	readout	57 percent excitatory, 43 percent inhibitory
Dopaminergic to output integration	Aso 2014	readout	301 dopaminergic, 0.28 of outputs respond
Kenyon to output convergence	Litwin-Kumar 2017	readout	median 13 Kenyon cells per output
Human disease genetics			
Disease-gene neuronal enrichment	Li 2022	readout	epilepsy 11.3 percent, z = 13.2
Ortholog rate in neuron genes	Hu 2011	readout	0.29 genome against 0.50 neuron-enriched
Neuropathy sensorimotor	Li 2022	readout	sensory z = 7.7, motor z = 6.7
Fly disease-model census	Wangler 2017	readout	models across every disease category
Parkinson gene bridge	Greene 2003	readout	park to PRKN, Pink1 to PINK1
Optogenetic targets			
Aminergic receptor panel	Nadim 2014	readout	20.1 percent against 5.4 percent, z = 4.2
Mechanosensation bridge	Coste 2010	readout	Piezo and TrpA1 to human channels
Circadian clock bridge	Guo 2016	readout	five clock genes to sleep-disorder orthologs
Integrate-and-fire calibration	Kakaria 2017	validation	threshold at 25 synaptic events
Hodgkin-Huxley response	Hodgkin 1952	validation	silent below threshold, monotone f-I